

Relatório Técnico de Testes de Performance

Resumo Executivo

O projeto foi testado em quatro cenários com k6: Smoke, Load, Stress e Spike.

No cenário de CPU (/checkout/crypto), o teste de stress com 1000 VUs apresentou uma taxa de falha de 84,6% e tempos de resposta elevados, indicando que a capacidade máxima confiável está abaixo de 1000 usuários simultâneos.

No cenário de I/O (/checkout/simple), a API suportou 50 VUs durante 2 minutos e picos de 300 VUs em spike test, sem falhas críticas significativas.

Análise de Estresse

O ponto de ruptura do cenário CPU começa a ser visível antes dos 1000 VUs.

No resultado do stress test, o número de requisições com resposta 201 caiu para 15%, e a taxa de erro subiu para 84,6%, mostrando que a aplicação já estava sobrecarregada. Recomenda-se investigar a capacidade entre 200 e 500 VUs para determinar o limite de confiança da API em operações CPU-heavy.

Evidências de Execução

Smoke Test

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PS C:\Users\jmd\Documents\caldwell\teste de software\teste de desempenho-rodrigo-carvalho\caldwell> java -jar smoke-test

> teste-de-desempenho -n 100 smoke-test
> 30 ms teste/smoke.js



execution local
script: tests/smoke.js
output:

scenarios: (100,005) 1 scenario, 1 max VMs, 100s max duration (incl. graceful stop):
    * default: 1 looping VM for 30s (gracefulstop: 30s)

running (00m30.3s), 4/1 VMs, 30 complete and 0 interrupted iterations
default ✓ (-----) 1 VMs 30s

✓ health endpoint returns 200

✓ checks:..... 100,005 / 30 0 / 0
data_received..... 8.2 GB 100 s/s
data_sent..... 2.6 MB 0 s/s
http_req_blocked..... avg=0.1ms min=0s max=0s p(90)=0s p(99)=0s
http_req_connecting..... avg=0.1ms min=0s max=0s p(90)=0s p(99)=0s
http_req_duration..... avg=1.76ms min=0.001s max=18.22ms p(90)=2.45ms p(99)=2.25ms
    expected response time..... avg=1.76ms min=0.001s max=18.22ms p(90)=2.45ms p(99)=2.25ms
✓ http_req_failed..... 0 / 30 0 / 0
http_req_receiving..... avg=0.11ms min=0s max=1.30ms p(90)=0.43ms p(99)=0.54ms
http_req_sending..... avg=0.27ms min=0s max=0.74ms p(90)=0.4ms p(99)=0.53ms
http_req_tls_handshaking..... avg=0.1ms min=0s max=0s p(90)=0s p(99)=0s
http_req_waiting..... avg=1.63ms min=0s max=17.64ms p(90)=2.33ms p(99)=2.47ms
http_req..... 30 0 0ms/1s
iteration_duration..... avg=1.661s min=1.615s max=1.825s p(90)=1.805s p(99)=1.825s
iterations..... 30 0 0ms/1s
vm..... 1 min=1 max=1

```

Load Test

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C:\Users\jrodriguez\Documents\Visual\Básico de software\Básico de desarrollo\redrig\carroll\hividad\open load-test
> teste de-desempenho,não finalizado
> cd run tests/load.js



execution: local
script: tests/load.js
output:

scenario: (100.00%) 1 scenario, 10 test WPs, 4444 test duration (incl. graceful steps):
    * default (up to 50 looping WPs for 3444s over 3 stages (gracefulTimeout: 30s, gracefulStop: 30s))

running (4444.3s), 00/50 WPs, 39996 complete and 0 interrupted iterations
default // [scenario: (100.00%) 1 scenario, 10 test WPs, 4444 test duration]

✓ status is pass

checks ..... 100.00% / 10000  0
data_received ..... 12.1B / 10000
data_sent ..... 7.4 MB / 10000
http_req_blocked ..... avg=0.00s min=0s max=0s max=0.00s p[99]=0s p[99]=0s
http_req_connecting ..... avg=0.00s min=0s max=0s max=0.00s p[99]=0s p[99]=0s
✓ http_req_duration ..... avg=0.00s min=0.00s max=0.00s max=0.00s p[99]=0.00s p[99]=0.00s
  ( expected response time ) 0.00s  0
http_req_failed ..... avg=0.00s min=0s max=0s max=0.00s p[99]=0.00s p[99]=0.00s
http_req_receiving ..... avg=0.00s min=0s max=0s max=0.00s p[99]=0.00s p[99]=0.00s
http_req_sending ..... avg=0.00s min=0s max=0s max=0.00s p[99]=0.00s p[99]=0.00s
http_req_tls_handshaking ..... avg=0.00s min=0s max=0s max=0.00s p[99]=0s p[99]=0s
http_req_waiting ..... avg=0.00s min=0.00s max=0.00s max=0.00s p[99]=0.00s p[99]=0.00s
iteration_duration ..... avg=0.00s min=0.00s max=0.00s max=0.00s p[99]=0.00s p[99]=0.00s
iterations ..... 39999  0
wps ..... 1  0
wps_max ..... 1  0

```

Stress Test

[illegible]

Spike Test

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PS C:\Users\jldin\Documents\Faculdade\Teste de software\teste-de-desempenho-rodrigo-carvalho\atividades> npm run spike:test

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execution: local
script: tests/spike.js
output: -

scenarios: (100.00%) 1 scenario, 300 max VUs, 260ms max duration (incl. graceful stop):
  * default: up to 300 looping VUs for 1m50s over 4 stages (gracefulrampdown: 30s, gracefulstop: 30s)

running (1m50.2s), 000/300 VUs, 102925 complete and 0 interrupted iterations
default / [=====] 000/300 VUs 1m50s

 ✓ status is 200

checks: ..... 100.00% ✓ 102925 X 0
data received: ..... 11 MB 226.80/s
data sent: ..... 19 MB 175.80/s
http_req_blocked: ..... avg=5.13µs min=0s med=0s max=7.68ms p(90)=0s p(95)=0s
http_req_connecting: ..... avg=7.12µs min=0s med=0s max=3.48ms p(90)=0s p(95)=0s
✓ http_req_duration: ..... avg=266.41ms min=99.64ms med=266.37ms max=315.62ms p(90)=286.74ms p(95)=297.23ms
  [ expected response:true ]
✓ http_req_failed: ..... 0.00% X 0 X 102925
http_req_receiving: ..... avg=0.00µs min=0s med=0s max=0.93ms p(90)=0s p(95)=546.6µs
http_req_sending: ..... avg=15.37µs min=0s med=0s max=51.64ms p(90)=0s p(95)=0s
http_req_tls_handshaking: ..... avg=0s min=0s med=0s max=0s p(90)=0s p(95)=0s
http_req_waiting: ..... avg=206.76ms min=99.78ms med=206.51ms max=315.62ms p(90)=286.68ms p(95)=297.27ms
http_res: ..... 102925 313.82/348.6/s
iteration_duration: ..... avg=206.76ms min=99.78ms med=206.51ms max=315.62ms p(90)=286.91ms p(95)=297.51ms
iterations: ..... 102925 933.42/1587/s

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